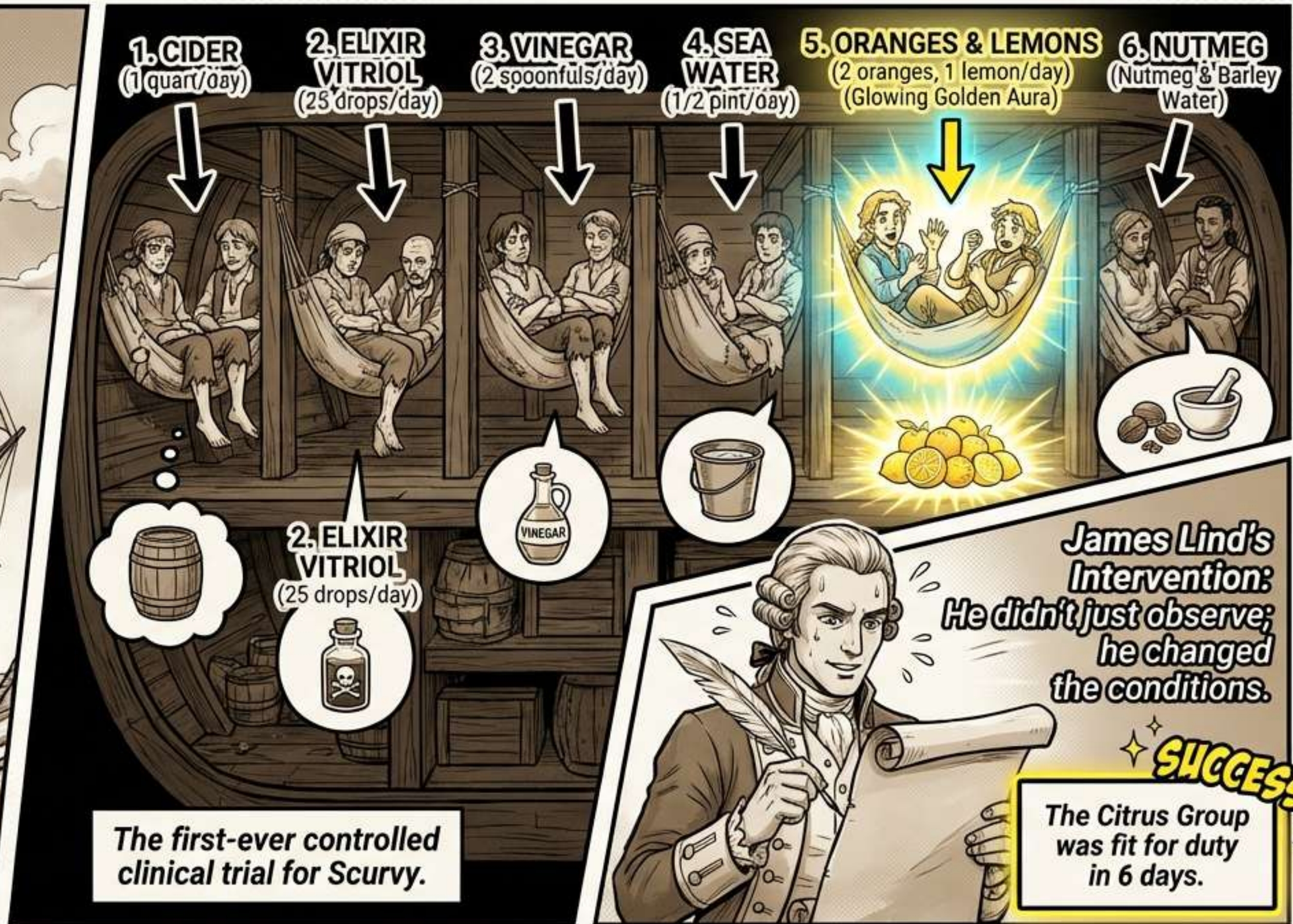


THE CHRONICLES OF CAUSALITY

MASTERY OF THE TRIAL: RCT, FIELD, & COMMUNITY EXPERIMENTS

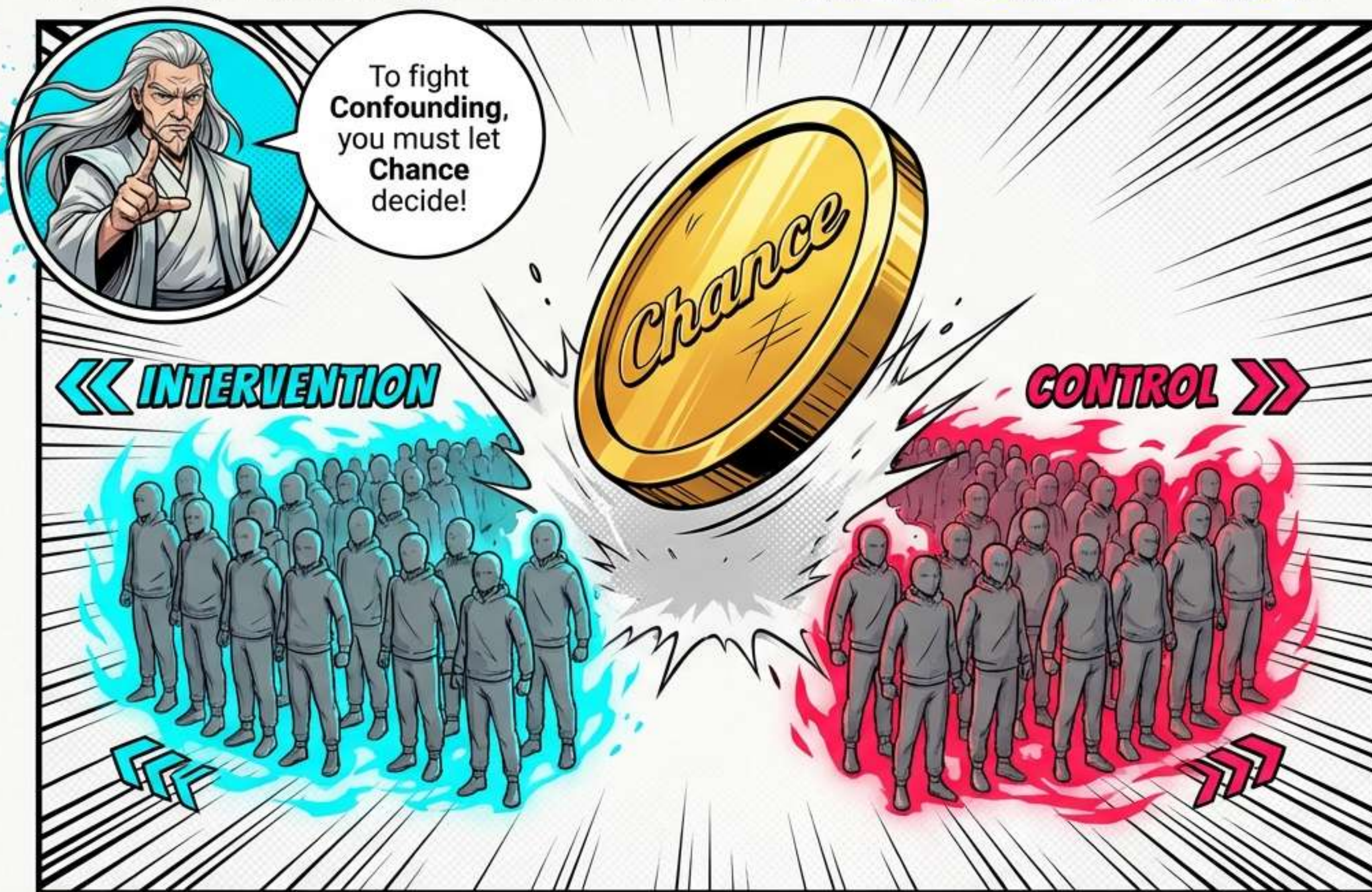


MISSION: To defeat the Monsters of Bias and Confounding, you must master the three legendary forms of experimentation. Only then can you prove Causality.



EXPERIMENTAL STUDIES: A study in which conditions are under the direct control of the investigator. We change the condition to find the cause.

THE ULTIMATE WEAPON: **RANDOMIZATION**



TACTICAL ADVANTAGES

1. **Eliminate Confounding**
(Balances prognostic factors).
2. **Create Comparable Groups.**
3. **Permit Probability Theory**
(Stats).

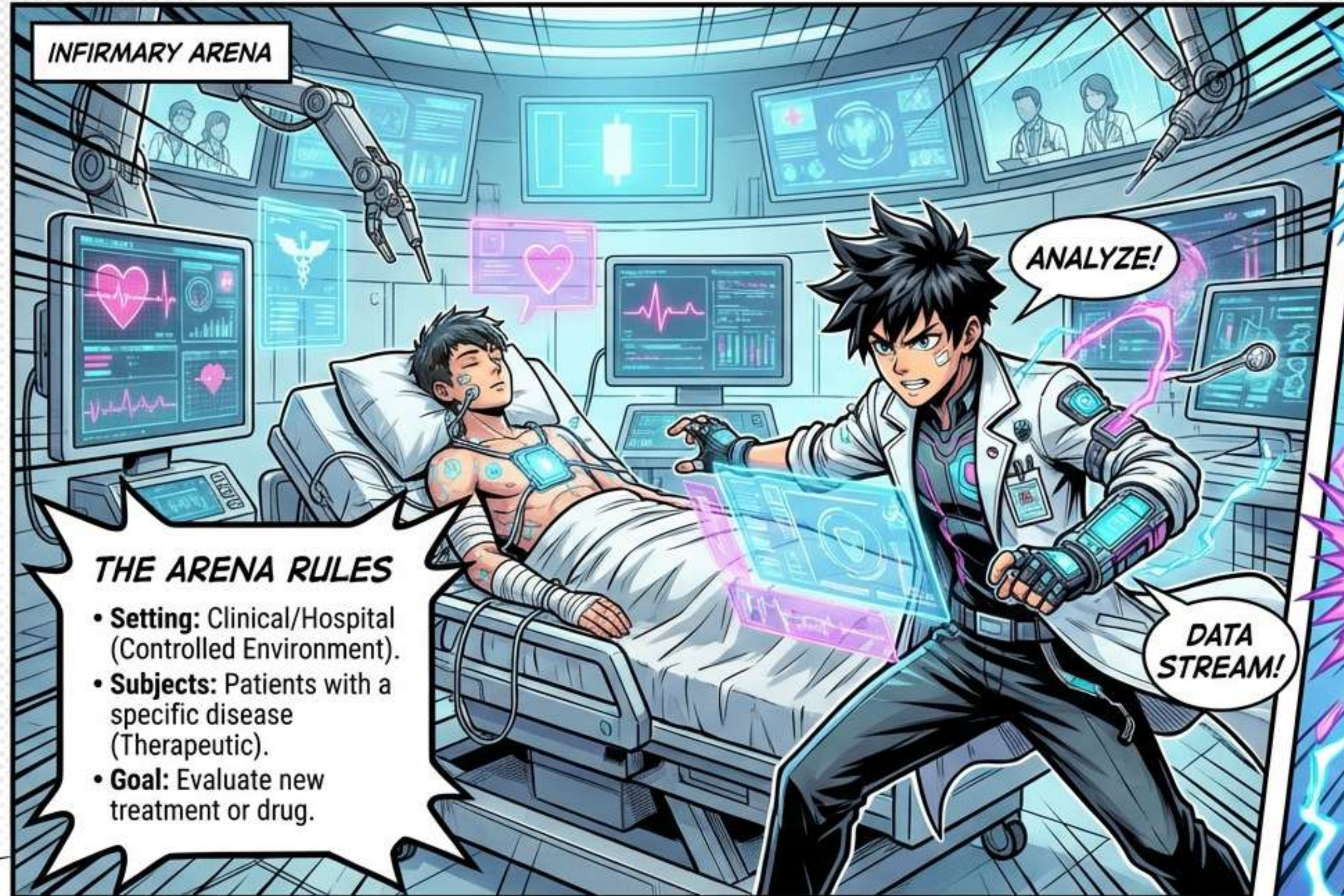
SECRET ITEM



The sequence must remain hidden until assignment.

EXPERIMENTAL STUDIES: A study in which conditions are under the direct control of the investigator. We change the condition to find the

ROUND 1: THE CLINICAL DUEL (RCT)



4 PHASES OF POWER



PHASE I: SAFETY & DOSAGE (Small Group)



Clinical patients shoit to assfety a dossage increasing analzing eoement, safety and smally and achumapoed inveimment.



PHASE II: EFFICACY & SAFETY



Efficacy patients with wrong enenies, medium group and sxafaty li enjoying with a standart, and nioriom.



PHASE III: CONFIRM EFFECTIVENESS vs STANDARD (Large Group)



Comparison-gomes are possie or large group of pere:csiooalty contincb-dave requiratv missements for the confirm and emirusness group.

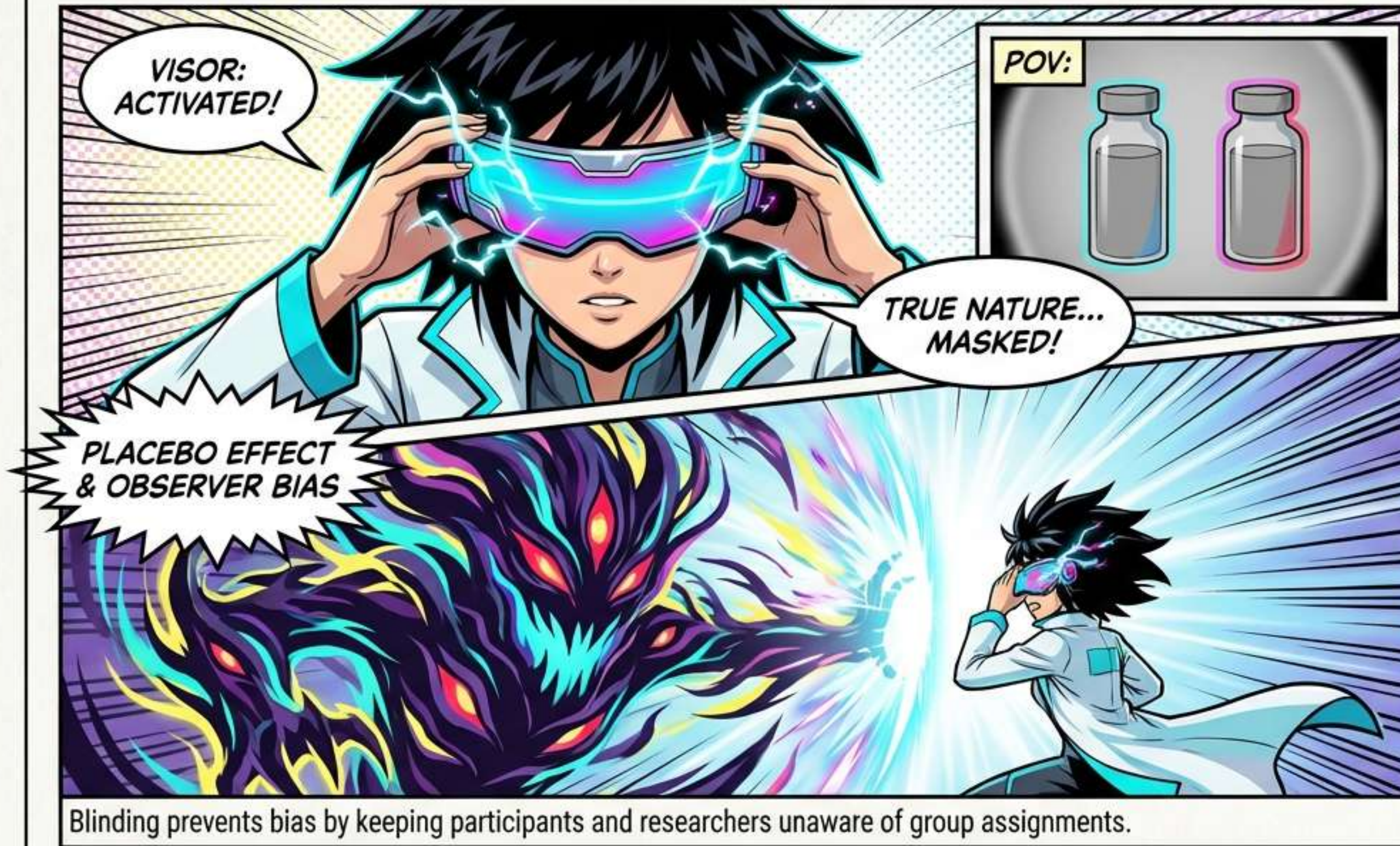


PHASE IV: POST-MARKETING SURVEILLANCE



Ongoing monitoring of promotion marketing stemoes thar:oveir targeting and corrolativns.

DEFENSE MOVE: BLINDING (MASKING)



1. SINGLE BLIND:
Subject is unaware.



2. DOUBLE BLIND:
Subject + Investigator are unaware (Gold Standard).



3. TRIPLE BLIND:
Subject + Investigator + Analyst are unaware.



ROUND 2: THE OPEN FIELD (FIELD TRIALS)



THE ARENA RULES

Setting: The Real World (Non-institutionalized).

Subjects: Healthy people at risk.

Goal: Prevention (Prophylactic) e.g., Vaccines.

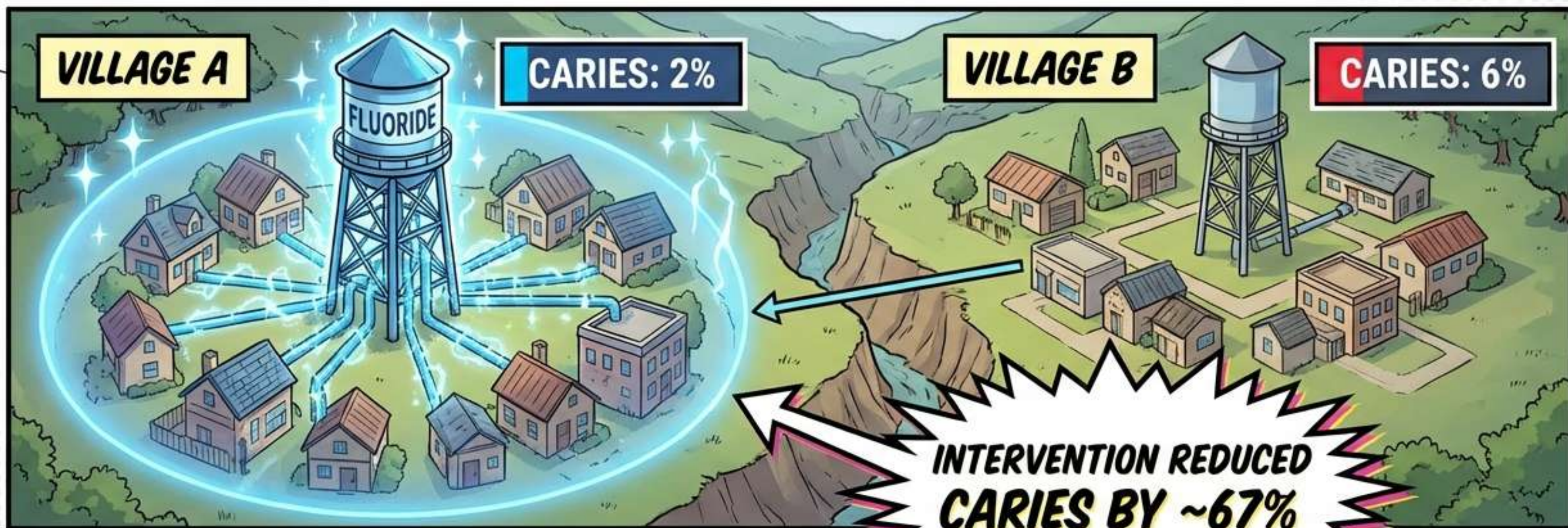
LOGISTICS:

High Cost.

Long Duration.

Huge Sample Size.

ROUND 3: THE VILLAGE DEFENSE (COMMUNITY TRIALS)



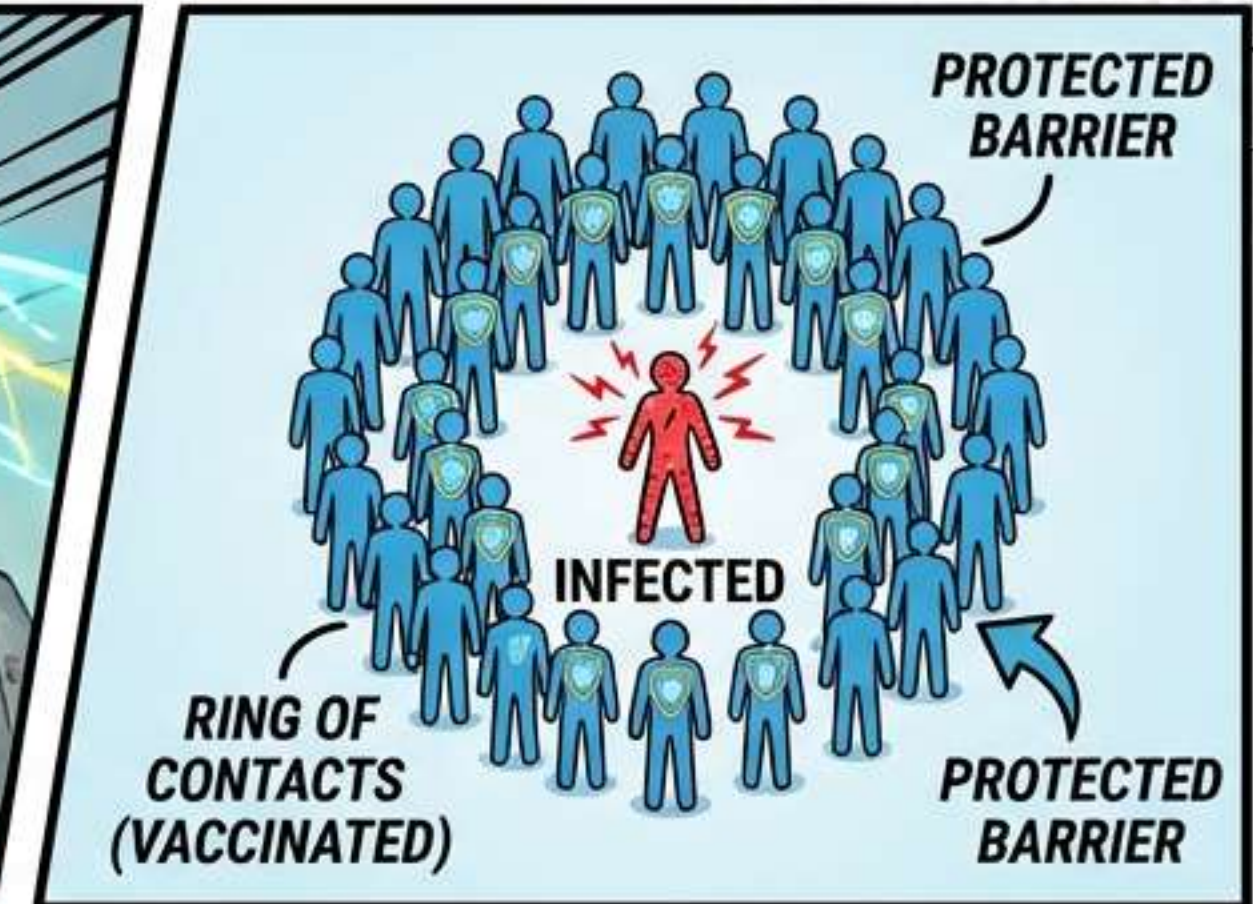
DEFINITION:

- ALLOCATION OF ENTIRE COMMUNITIES.

USE CASE:

- INTERVENTIONS THAT CANNOT BE SEPARATED BY INDIVIDUAL (WATER, AIR, MASS MEDIA).

THE CLUSTER COMPLICATION



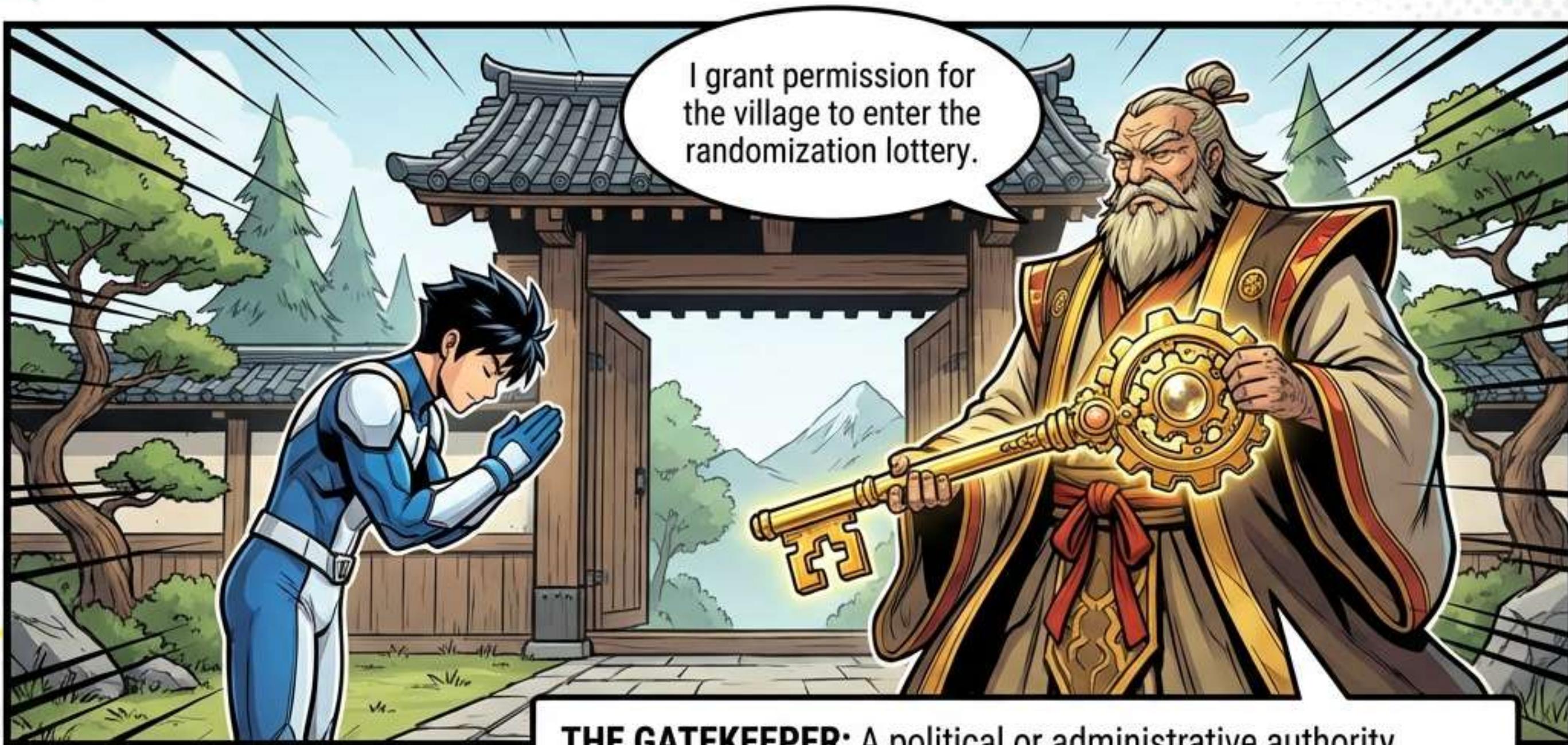
THE PROBLEM: Contamination & Herd Immunity.

THE SOLUTION: Cluster Randomization (Randomize the Village, not the Person).

EXAMPLE: Ebola Ring Vaccination (Guinea).



GUARDIANS OF THE CLUSTER: THE GATEKEEPER

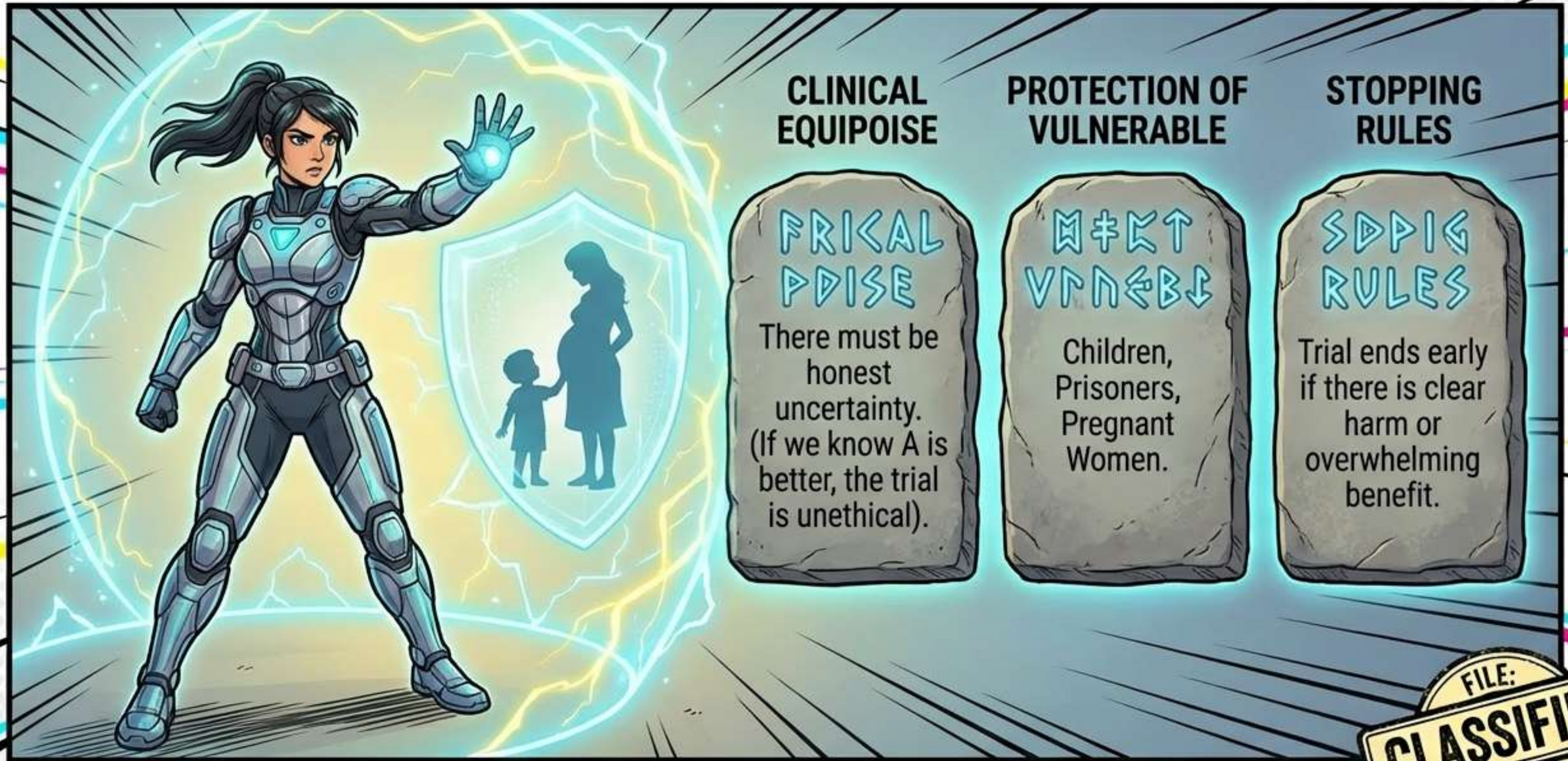


THE GATEKEEPER: A political or administrative authority (Mayor, Headmistress).

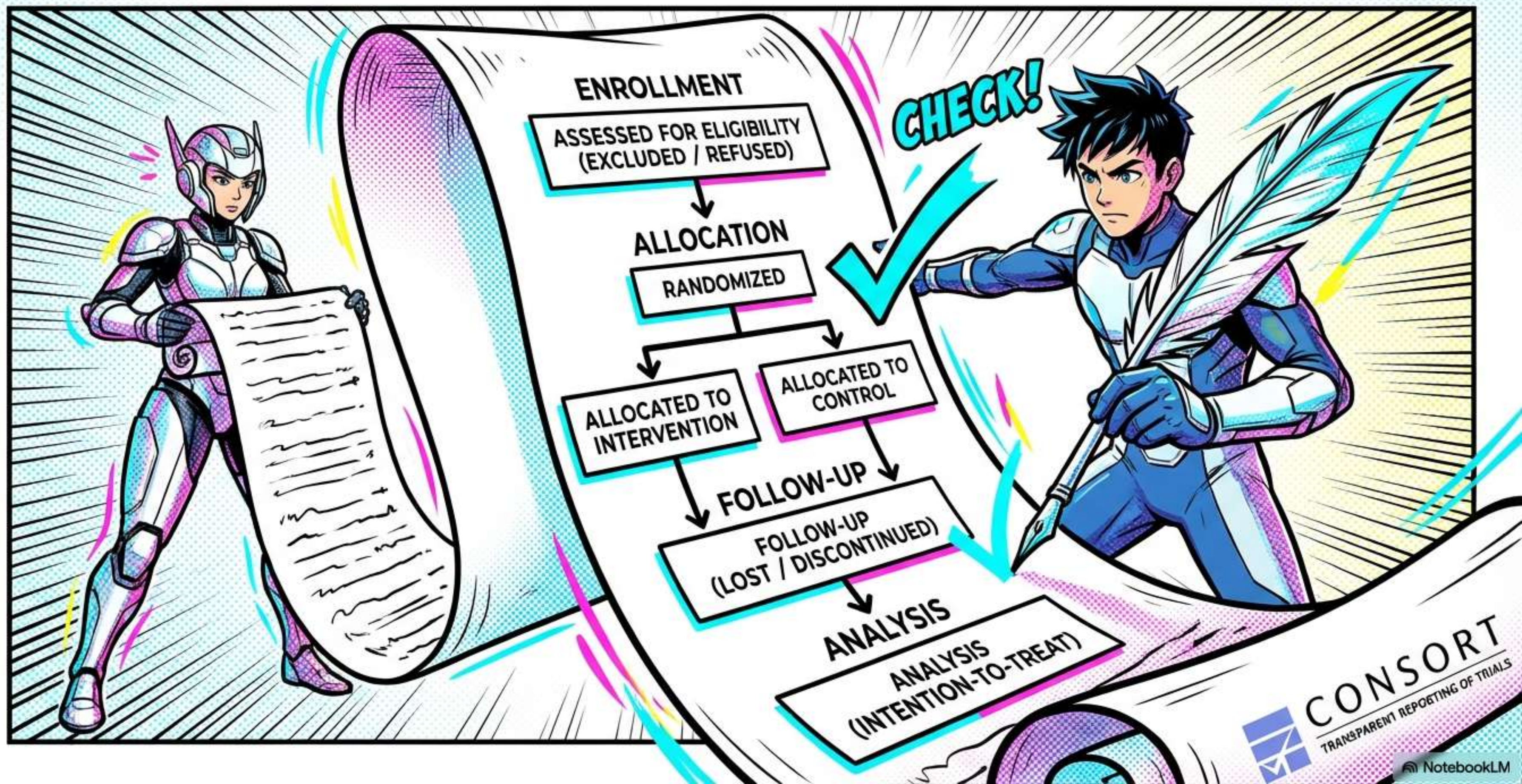
THE RULE: Gatekeeper permission is the *first step*, but individual consent is still required where feasible.

REFERENCE: "The Ottawa Statement."

THE CODE OF HONOR: ETHICS



THE SCROLL OF TRUTH: CONSORT



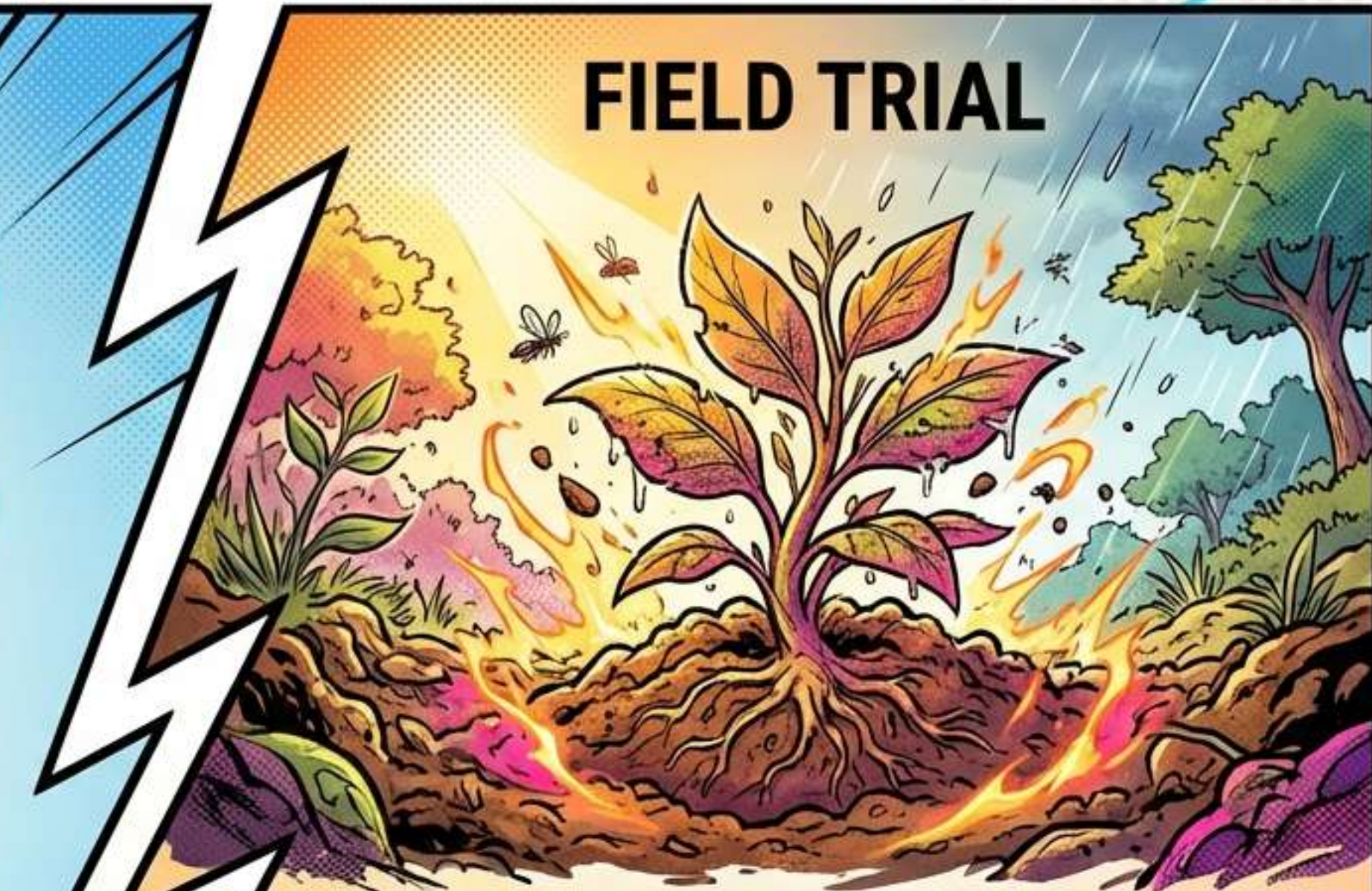
TACTICAL ANALYSIS: LAB VS. FIELD

CLINICAL RCT



- High Control (Internal Validity)
- **Artificial** Environment
- Best for Mechanisms

FIELD TRIAL



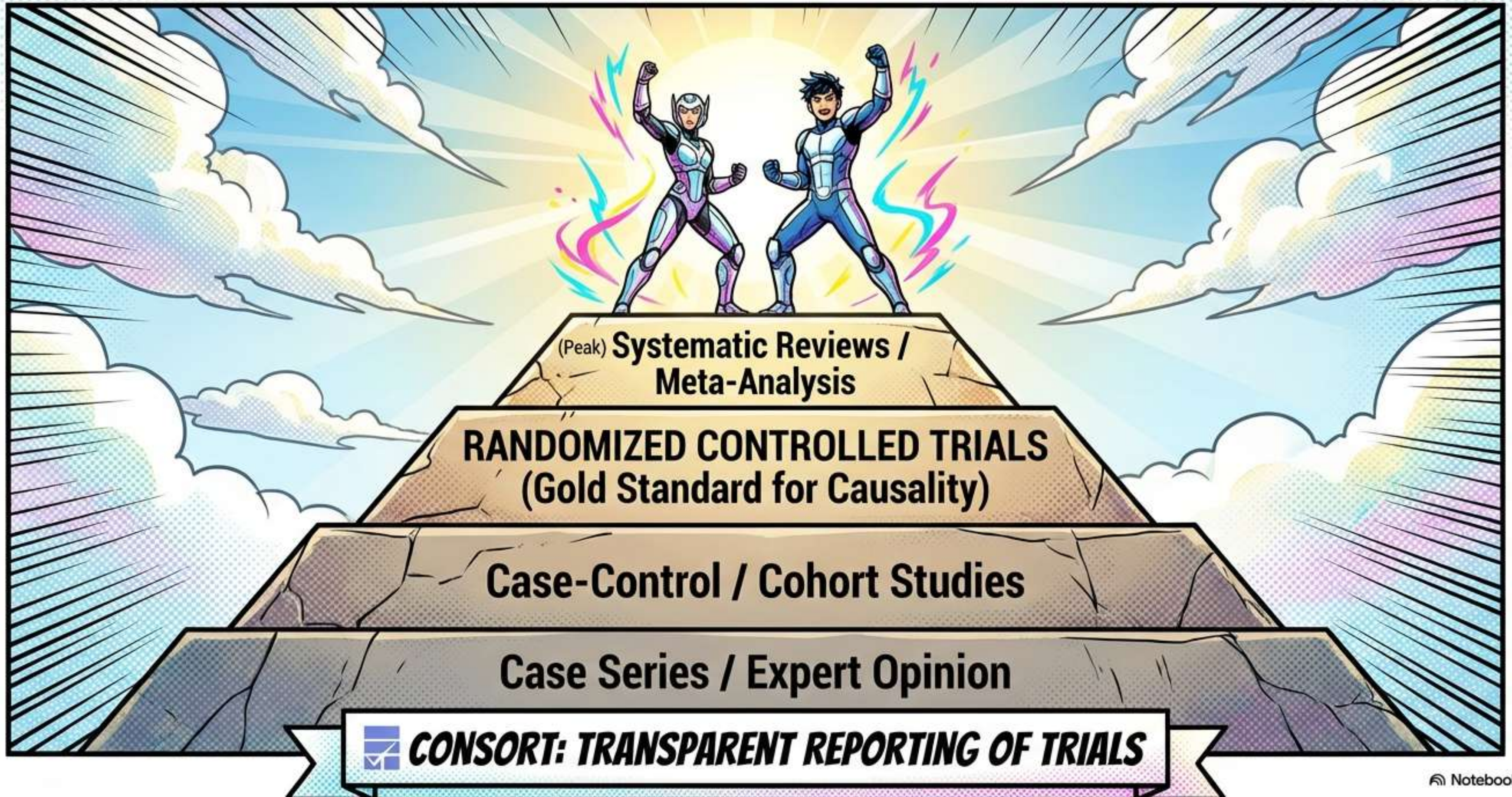
- High Realism (External Validity)
- Natural Environment
- Best for Real-World Policy

CONTROL

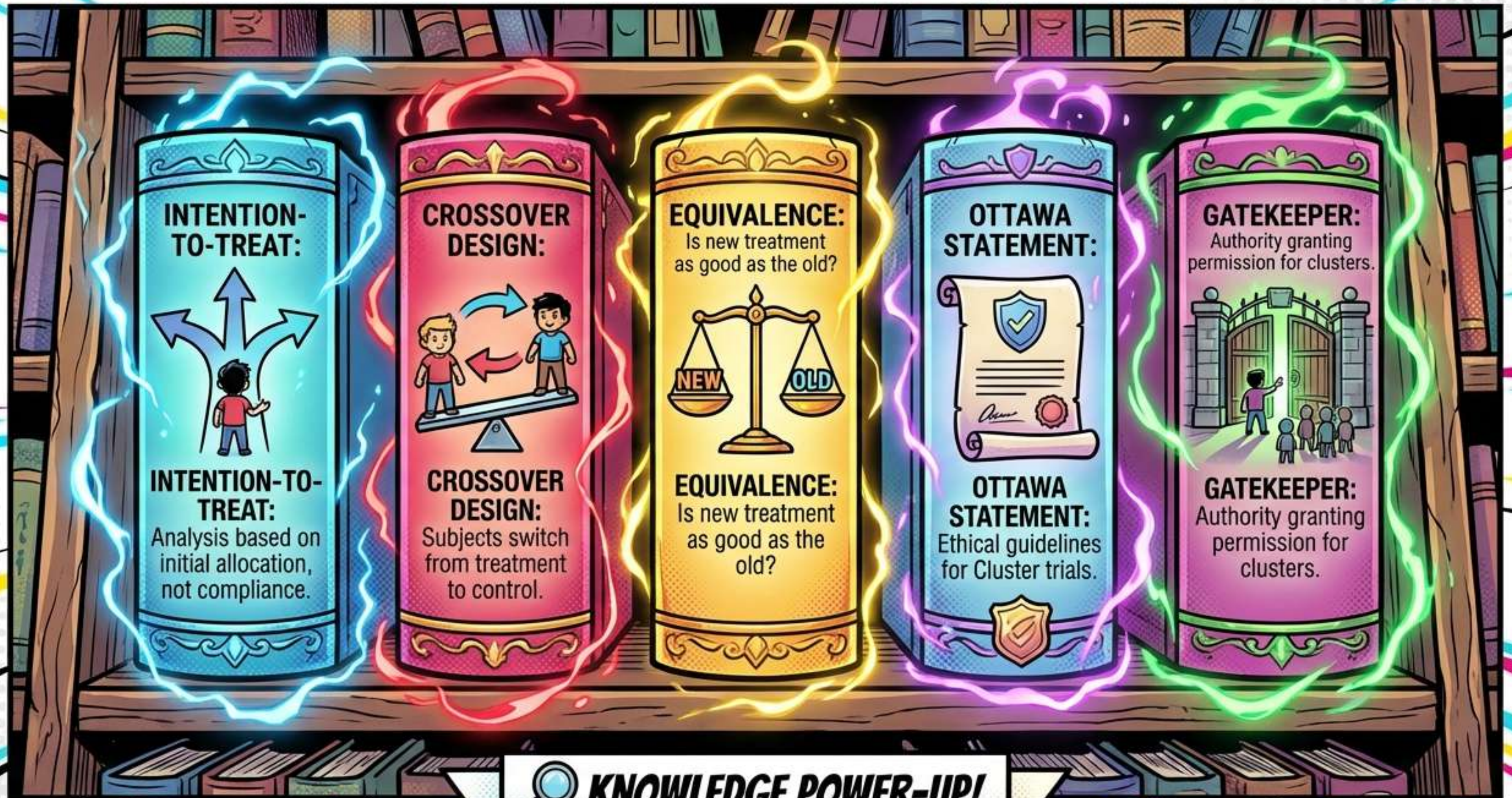
GENERALIZABILITY

FILE:
CLASSIFIED

MASTERY ACHIEVED: HIERARCHY OF EVIDENCE



THE LIBRARY OF KNOWLEDGE (GLOSSARY)



REFERENCES & SOURCES

- Gordis, L. (2014). **Epidemiology.**
- Weijer, C. et al. (2011). **Ethical issues posed by cluster randomized trials.**
- Coldiron, M. (WHO). **Cluster Randomized Trials in Health EDRM.**
- **CONSORT 2010 & 2025 Statements.**
- Bonita, R. et al. (2006). **Basic Epidemiology.**
- **Excellence in Breeding Platform (Field Trials).**

